

Performance Testing

Project Title: Rising Waters

Introduction

Performance testing is conducted to verify that the Rising Waters system performs efficiently under normal operating conditions. The testing focuses on prediction speed, accuracy, system stability, and user response.

Testing Process

The application was tested using different weather and rainfall input values. The trained Machine Learning model processed the data and generated prediction results through the Flask web application. Multiple test cases were executed to evaluate the overall system performance.

Performance Results

The system successfully accepted user inputs, processed the data, and displayed flood prediction results within a short time. The application remained stable during repeated testing and produced consistent prediction outputs based on the trained model.

Observations

The prediction process was fast and responsive. The web application loaded correctly, accepted valid inputs, and displayed appropriate flood or no-flood results. Error handling also worked properly when invalid data was entered.

Conclusion

The performance testing confirmed that the Rising Waters system is reliable, responsive, and capable of providing quick flood predictions. The application performs efficiently and is suitable for real-time flood risk prediction.